

Summary of communications to Fulton Hogan: (for the lecturers to answer)

- Does it make sense to plan roadworks for school holidays in CBD areas ?
- Which days are best if roadworks must be completed during the day ?
- Are there any geographical differences between CBDs we must be aware of ?

Noting that Fulton Hogan is investing in the consulting group's success, I would ask - "What are the metrics that are in line with FH operational and safety priorities?"

- Is the assignment to determine the numeric metrics for FH to make the decision? or,
- Is the assignment to determine if known numeric metrics are met for FH to make a decision?

If it is the second, can FH provide these values?

I have used a LLM to seek some ballpark numbers for strawman result. The following is what was obtained from ChatGPT.

Strawman/Example Thresholds for Decision-Making

1. Does it make sense to plan roadworks for school holidays in CBD areas?

- **Traffic Volume Reduction:**
 - **Threshold:** A reduction of at least 20–30% in average daily traffic volume during school holidays compared to regular periods could indicate that roadworks would be less disruptive.
- **Pedestrian and Business Activity:**
 - **Threshold:** A decrease of 25% or more in pedestrian foot traffic or business operations (e.g., reduced hours or closures) during school holidays could justify scheduling roadworks at these times.
- **Safety Incident Reduction:**
 - **Threshold:** A decrease of 15% or more in safety incidents (e.g., traffic accidents, worker injuries) during school holidays compared to regular periods may indicate safer conditions for roadworks.

2. Which days are best if roadworks must be completed during the day?

- **Optimal Traffic Flow:**
 - **Threshold:** Days with traffic volumes consistently below 60% of the weekly average could be considered ideal for daytime roadworks.
- **Business Activity:**
 - **Threshold:** Days when business activity (e.g., retail transactions, foot traffic) is 20% lower than the weekly average can be targeted to minimise disruption.
- **Safety Metrics:**
 - **Threshold:** Days with 10% or fewer recorded safety incidents compared to the average may be safer for conducting roadworks.

3. Are there any geographical differences between CBDs we must be aware of?

- **Traffic and Congestion Differences:**

- **Threshold:** If one city's CBD has traffic congestion 25% higher than others, more stringent traffic management and scheduling may be required.

- **Business and Economic Impact:**

- **Threshold:** If business operation levels differ by more than 15% between cities during school holidays, city-specific strategies should be considered.

- **Safety Incident Variations:**

- **Threshold:** If one city has a 20% higher rate of safety incidents during roadworks compared to others, additional safety measures should be implemented.

These thresholds are starting points that can help establish whether school holidays are more suitable for roadworks and how to optimise the timing and location of these projects. The specific thresholds should be refined based on historical data and stakeholder input to align with Fulton Hogan's operational and safety priorities.